

(1001CJA102120057)

Test Pattern

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Advanced)
FULL SYLLABUS
21-03-2021
JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE-II)**ANSWER KEY****PAPER-1****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B,D	A,B,C,D	A,C,D	A,B,C	A,C,D	A,B,C,D	A	A	A	B
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	4.80	2.82 to 2.83	60.00	10.38 to 10.40	1.84	0.00	6.00	6.00		

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B,C	B,C,D	B,C,D	B,D	A,C,D	A,B,C	B	C	B	D
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	23.72	5.00	0.10	13.00	6.00	28.00 to 34.00	6.00	56.00		

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B,C,D	B,C	A,B,C,D	A,C,D	A,B,D	A,C,D	A	C	C	B
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	27.00	6.82 or 6.83	13.66 or 13.67	7.00	80.00	2.00	16.00	7.00		

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JEE(Advanced)
FULL SYLLA-
21-03-2021
JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE-II)**ANSWER KEY****PAPER-2****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,D	B,C	A,B,D	A,B,C	A,D	A,D	B	A	A	B
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	0.25	12.00	6.00	1.33	2.00	6.66 to 6.67	3.00	17.00		

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B,C	B,C	B	A,C	A,B,D	A,B,C,D	B	C	B	C
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	5.66	6.00	9.10 TO 9.40	36.00	6.00	0.18 TO 0.20	4.00	5.00		

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,C	A,C,D	B,C	B,D	A,C	B,C	A	C	A	A
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	1.00	4.00	1.00	6.00	8.00	10.00	3.00	10.00		

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE-II)
PAPER-1
PART-1 : PHYSICS
SOLUTION
SECTION-I

 1. **Ans. (A,B,D)**

Sol. Activity of A = λN_A
 Activity of B = $2\lambda N_B$
 $\lambda N_A = 2\lambda N_B$
 $N_A = 2N_B$

 2. **Ans. (A,B,C,D)**

Sol. $w = \frac{\lambda D}{d}$
 $n = \left(\frac{xd}{\lambda D} \right)$
 $\frac{dn}{dt} = \left(\frac{xv}{\lambda D} \right)$

 3. **Ans. (A,C,D)**

Sol. net force on magnet is zero in uniform magnetic field.

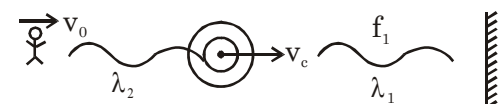
 4. **Ans. (A,B,C)**

Sol. $a = -bx = -w^2x$ $w = \sqrt{b}$
 $V_{\max} = aw$ $A = \frac{u}{\sqrt{b}}$

 5. **Ans. (A,C,D)**

Sol. $H_{\max} = \frac{2T \cos \theta}{r \rho g} = 1 \text{ cm}$
 $H \propto \cos \theta_C$

 6. **Ans. (A,B,C,D)**

Sol. 

$$\lambda_1 = (v - v_c)T = \left(\frac{v - v_c}{f} \right)$$

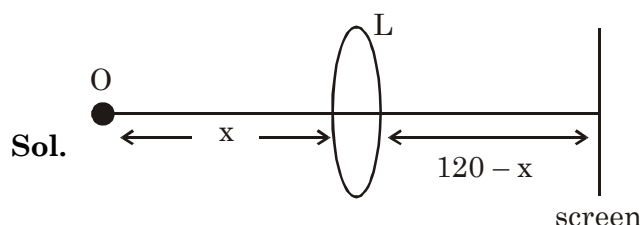
$$\lambda_2 = (v + v_c)T = \left(\frac{v + v_c}{f} \right)$$

$$f_1 = \frac{v}{x_1} = \left(\frac{v}{v + v_c} \right) f$$

$$\Delta f_1 = f_{\text{echo}} - f_{\text{street}}$$

$$= \left(\frac{v}{v - v_c} \right) f \cdot \left(\frac{v + v_1}{v} \right) - f \left(\frac{v + v_0}{v + v_c} \right)$$

$$= \frac{2v_c (v + v_0) f}{(v^2 - v_c^2)}$$

 7. **Ans. (A)**


$$m_1 : m_2 = 1 : 9$$

$$m_1 \cdot m_2 = 1$$

$$\therefore m_1 = \frac{1}{3} \quad m_2 = 3$$

$$m = \frac{v}{u}$$

$$\frac{1}{3} = \frac{120 - x_1}{x_1}$$

$$4x_1 = 360$$

$$x_1 = 90$$

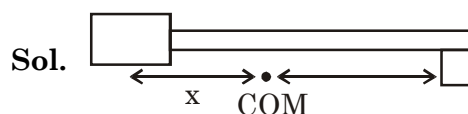
$$x_2 = 120 - 90 = 30$$

$$D = 120$$

$$d = x_1 - x_2 = 60$$

$$f = \frac{D^2 - d^2}{4D} = 22.5 \text{ cm}$$

 8. **Ans. (A)**

 9. **Ans. (A)**


$$x = \frac{M.L.}{3M + M} = \frac{L}{4}$$

from conservation of angular momentum about COM

$$Mv_0 \left(\frac{3L}{4} \right) = Iw$$

$$I = 3M \left(\frac{L}{4} \right)^2 + M \left(\frac{3L}{4} \right)^2 = \frac{3ML^2}{16} + \frac{9ML^2}{16}$$

$$= \frac{3}{4} ML^2$$

$$\frac{3Mv_0 L}{4} = \frac{3}{4} ML^2 \cdot w$$

$$\boxed{w = \frac{v_0}{L}}$$

10. Ans. (B)

Sol. From cons of linear momentum

$$Mv_0 = 4M \cdot v_c$$

$$v_c = \frac{v_0}{4}$$

$$\text{Velocity of } 3M = v_c - \frac{L}{4} w$$

$$= \frac{v_0}{4} - \frac{L}{4} \times \frac{v_0}{L} = 0$$

SECTION-II

1. Ans. 4.80

$$\text{Sol. } \frac{1}{t} = \frac{1}{t_1} + \frac{1}{t_2}$$

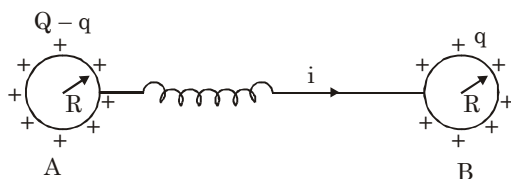
$$t = \frac{t_1 t_2}{t_1 + t_2} = \frac{24 \times 6}{24 + 6} = 4.80 \text{ Hr.}$$

2. Ans. 2.82 to 2.83

Sol. Let q = charge on B at time ' t '

$$V_A - V_B = L \frac{di}{dt}$$

$$\Rightarrow K \frac{Q-q}{R} - K \frac{q}{R} = L \frac{d^2 q}{dt^2}$$



[Spheres are at large distance. Hence charge on one does not affect the potential of other]

$$\Rightarrow \frac{d^2 q}{dt^2} = -\frac{2K}{LR} \left[q - \frac{Q}{2} \right] \dots\dots (i)$$

$$\text{Let } q - \frac{Q}{2} = x \text{ which means } \frac{d^2 q}{dt^2} = \frac{d^2 x}{dt^2}$$

In terms of x , the differential equation (i) becomes

$$\frac{d^2 x}{dt^2} = -\frac{2K}{LR} x$$

Solution to this equation is

$$x = x_0 \sin(\omega t + \delta) \quad \left[\omega = \sqrt{\frac{2K}{LR}} \right]$$

$$\Rightarrow q - \frac{Q}{2} = x_0 \sin(\omega t + \delta)$$

$$\Rightarrow q = \frac{Q}{2} + x_0 \sin(\omega t + \delta)$$

Current

$$i = \frac{dq}{dt} = x_0 \omega \cos(\omega t + \delta)$$

Just after closing the switch (at $t = 0^+$) the current is zero

$$\therefore \delta = \frac{\pi}{2}$$

$$\therefore q = \frac{Q}{2} + x_0 \cos \omega t$$

Also, at $t = 0$, $q = 0$

$$\therefore x_0 = -Q/2$$

$$\therefore q = \frac{Q}{2} [1 - \cos \omega t]$$

$$q = \frac{Q}{2} \text{ means } \cos \omega t = 0$$

$$\Rightarrow \omega t = \frac{\pi}{2}$$

$$\Rightarrow \sqrt{\frac{2K}{LR}} t = \frac{\pi}{2}$$

$$\Rightarrow t = \frac{\pi}{2} \sqrt{\frac{LR}{2K}} = \frac{\pi}{2\sqrt{2}} = \sqrt{\frac{LR}{K}}$$

$$\begin{aligned} \therefore a &= 2\sqrt{2} \\ &= 2 \times 1.414 \\ &= 2.828 \\ &\approx 2.83 \end{aligned}$$

3. Ans. 60.00

Sol. Initially $V_R = V_{C_1} = 2.5 \times 4 = 10V$

$$Q_1 = CV = 3 \times 10 = 30 \mu C$$

$$U_1 = \frac{1}{2} \times 3 \times 10^2 = 150 \mu J$$

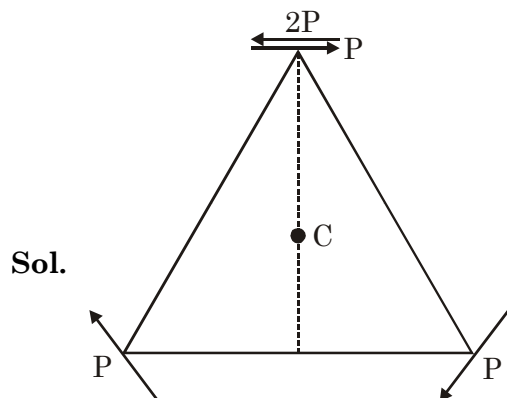
$$\text{Finally } V_{C_1} = V_{C_2} = \frac{Q_1}{C_1 + C_2} = \frac{30}{5} = 6 V$$

$$U_2 = \frac{1}{2} \times (C_1 + C_2) \times V^2$$

$$= \frac{1}{2} \times 5 \times 36 = 90 \mu J$$

$$\therefore \text{Heat liberated} = 150 - 90 = 60 \mu J$$

4. Ans. 10.38 to 10.40



Sol.

From superposition

$$E = \left(\frac{2KP}{r^3} \right) \text{ opposite to } 2\vec{P}$$

$$= \frac{2KP}{\left(\frac{L}{\sqrt{3}} \right)^3} \quad r = \frac{2}{3} \times \frac{\sqrt{3}L}{2} = \frac{L}{\sqrt{3}}$$

$$= 6\sqrt{3} \left(\frac{KP}{L^3} \right)$$

$$\therefore a = 6 \times 1.732 = 10.392$$

5. Ans. 1.84

Sol. Length of 1 main scale division

$$= 1/10 = 0.1 \text{ cm}$$

No. of vernier scale divisions = 10

Main scale reading = 1.8 cm

Coinciding vernier scale division = 4th

Least Count

= Length of smallest main scale division/

No. of vernier scale divisions

$$= 0.1 / 10$$

$$= 0.01 \text{ cm}$$

(vernier calliper has a least count 0.01 cm. So, measurement is accurate only upto three significant figures.)

length = Main scale reading + coinciding vernier scale division $\times$ LC

$$= 1.8 + 4 \times 0.01$$

$$= 1.8 + 0.04$$

$$= 1.84 \text{ cm}$$

6. Ans. 0.00

Sol. Heat lost by water = Heat given by ice

$$(3 + 1) \times 1 \times (20 - \theta)$$

$$= 1 \times 0.5 \times 40 + 1 \times 80 + 1 \times 1 \times \theta$$

$$80 - 4\theta = 20 + 80 + \theta$$

$$5\theta = 20$$

$$\text{Not possible} \Rightarrow \theta = 0^\circ$$

7. Ans. 6.00

Sol. After every collision velocity becomes

$$\frac{1}{\sqrt{2}} \text{ times.}$$

$$\frac{1}{8} = \left(\frac{1}{\sqrt{2}} \right)^n \Rightarrow n = 6$$

8. Ans. 6.00

Sol. Magnitude of viscous force, $F = \eta A \frac{dv}{dr}$

$$\Rightarrow \text{viscous force per unit area } \sigma = \frac{F}{A} = \eta \frac{dv}{dr}$$

$$v = v_0 \left(1 - \frac{r^2}{R^2} \right)$$

$$\Rightarrow \frac{dv}{dr} = -\frac{2v_0 r}{R^2}$$

$$\Rightarrow \sigma = \eta \frac{2v_0 r}{R^2} \quad \dots\dots (i)$$

Volume rate of flow, Q

consider an annular element at r from axis, width dr.

$$dA = 2\pi r dr; \quad dQ = v \cdot dA = v_0 \left(1 - \frac{r^2}{R^2} \right) 2\pi r dr$$

$$Q = \int dQ = 2\pi v_0 \left[\frac{r^2}{2} - \frac{r^4}{4R^2} \right]_0^R = \frac{\pi}{2} R^2 v_0$$

$$\Rightarrow v_0 = \frac{2Q}{\pi R^2}$$

$$\therefore (i) \Rightarrow \sigma = \eta \frac{4Q}{\pi R^4} r, \quad R = 0.1 \text{ m}$$

At r = 0.04 m,

$$\sigma = (0.75) 4 \times \frac{\pi}{2} \times 10^{-2} \times \frac{0.04}{\pi \times 10^{-4}}$$

$$= 6 \text{ Nm}^{-2}$$

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. **Ans. (A,B,C)**

Sol. (A) Only first four spectral lines belonging to Balmer series in hydrogen spectrum lie in visible region.

(B) If a light of frequency ν falls on a metal surface having work function $h\nu_0$, photoelectric effect will take place only if $\nu \geq \nu_0$, since ν_0 is the minimum frequency required for photoelectric effect.

(C) The number of photoelectrons ejected from a metal surface in photoelectric effect depends upon the intensity of incident radiations.

2. **Ans. (B,C,D)**

Sol. B : Magnesite is ore of Mg

C : Anode mud will have Au only

D : Self reduction occurs during roasting.

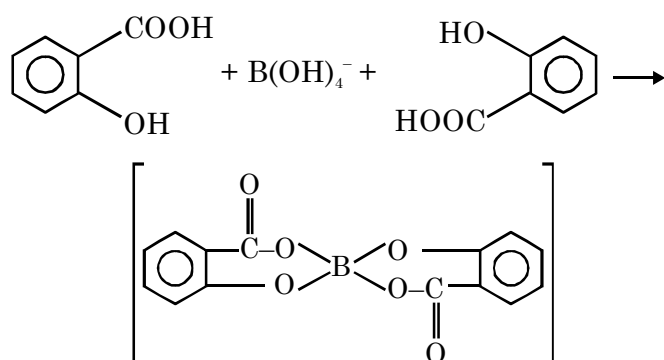
3. **Ans. (B,C,D)**

4. **Ans. (B,D)**

5. **Ans. (A,C,D)**

Sol. $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O} + \text{aq} \rightleftharpoons 4\text{H}_2\text{BO}_3 + 2\text{NaOH}$;

$\text{H}_2\text{BO}_3 + \text{OH}^- \rightleftharpoons \text{B(OH)}_4^-$



– Optically resolvable due to asymmetric structure

– Two six membered rings.

6. **Ans. (A,B,C)**

Sol. Option (D) incorrect because

• Rate of SN^1 depend on stability of carbocation formed.

7. **Ans. (B)**

Sol. $\text{A} + \text{B} \rightarrow \text{C} + \text{D}$

$t=0$ 1 1 0 0

$t=t$ 0.25 0.25 0.75 0.75

$$\frac{dx}{dt} = k(a-x)^{1/2} (b-x)^{1/2}$$

Here $a = b$

$$\frac{dx}{dt} = k(a-x)$$

$$\therefore t = \frac{1}{k} \ln \frac{a}{a-x}$$

$\therefore$ We'll treat it as first order, single reactant.

$$\therefore t = 2 \times t_{1/2} = 2 \times 0.693 / 2.31 \times 10^{-3} = 600 \text{ s.}$$

8. **Ans. (C)**

$$\text{Sol. } \frac{d}{dT}(\ln k) = \frac{E_a}{RT^2}$$

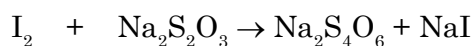
$$0 + \frac{2500}{T^2} + \frac{3}{T} = \frac{E_a}{RT^2}$$

$$\therefore E_a = (2500 + 3T)R$$

9. **Ans. (B)**



white

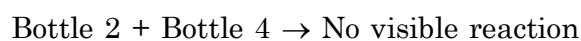


reduced oxidised

10. **Ans. (D)**

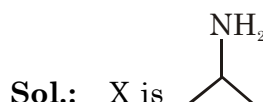
Sol. Bottles 1, 2, 3, 4 have $\text{Pb}(\text{NO}_3)_2$, HCl , Na_2CO_3 and CuSO_4 .

When mixing :



SECTION-II

1. **Ans. (23.72)**



$$\% \text{ Nitrogen} = \frac{14}{59} \times 100 = 23.72\%$$

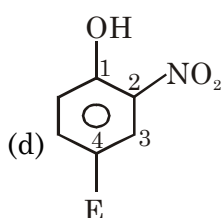
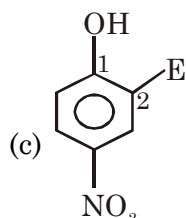
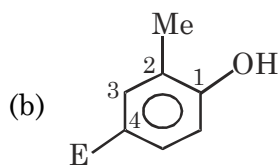
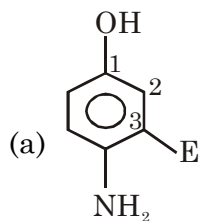
2. **Ans. (5.00)**

Sol.: 1, 4, 5, 6, 8 can be oxidised.

3. **Ans. (0.10)**

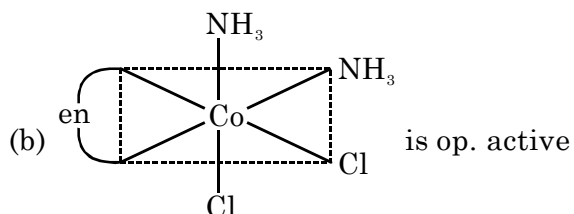
4. Ans. (13.00)

Sol.: $3 + 4 + 2 + 4 = 13$



5. Ans. (6.00)

Sol.: (a) Cis-isomer is op. active



(c) optically inactive.

(d) op. inactive

(e) always optically active

(f) square planar, optically inactive

(g) EDTA complexes are optically active

(h) 3 didentate ligands are always optically active

(i) same as (h)

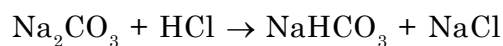
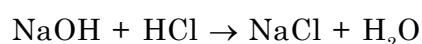
6. Ans. (28.00 to 34.00)

Sol.: $\frac{(\text{In}^-)}{(\text{HIn}^-)} = \frac{25}{75} = \frac{1}{3}$

$$\text{pH} = 9.6 + \log \frac{(\text{In}^-)}{(\text{HIn}^-)}$$

$$= 9.6 + \log \frac{1}{3}$$

$$\text{pH} = 9.12 = x$$



$$\text{pH} = \text{pK}_{a2} + \log \frac{(\text{CO}_3^{2-})}{(\text{HCO}_3^-)}$$

$$9.12 = 11 + \log \frac{1-x}{x}$$

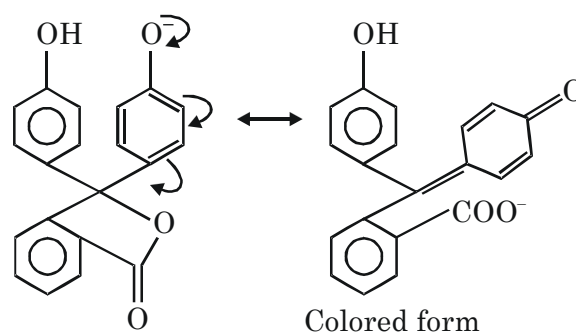
$$1.88 = \log \frac{x}{1-x}$$

$$75.85 = \frac{x}{1-x}$$

$$x = \frac{75.85}{76.85} = 0.99$$

$$\therefore 0.99 = 0.1 \times V_{\text{HCl}}$$

$$V_{\text{HCl Total}} = 9.9 + 10 = 19.9 \text{ ml} = y$$



$$\therefore z = 0$$

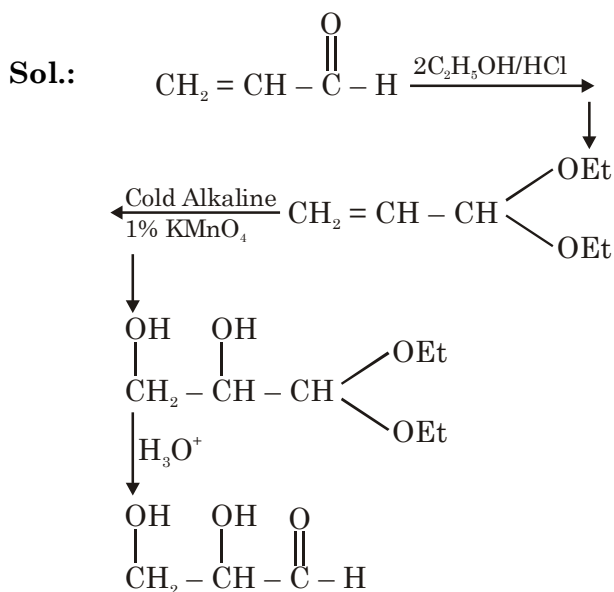
$$\therefore x + y + z = 9.12 + 19.9 + 0$$

$$= 29.02$$

7. Ans. (6.00)

Sol.: a, b, c, f, h, i

8. Ans. (56.00)



PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. Ans. (B,C,D)

Sol. $\Delta = \begin{vmatrix} p & q & r \\ q & r & p \\ r & p & q \end{vmatrix} = 0$ (for Non trivial solution)

$$-(p+q+r)(p^2+q^2+r^2-pq-qr-rp) = 0$$

$$p+q+r = 0$$

Represents line $x = y = z$

$$p^2+q^2+r^2-pq-qr-rp = 0$$

$$\frac{1}{2}(p-q)^2 + (q-r)^2 + (r-p)^2 = 0$$

p, q, r must be equal $p = q = r$

$\Rightarrow$ identical planes $x + y + z = 0$

2. Ans. (B,C)

Sol. $-2 < \log_{\frac{1}{\sqrt{3}}} \left(\frac{|z|^2 - |z| + 1}{2 + |z|} \right) < 2$

$$\left(\frac{1}{\sqrt{3}} \right)^2 < \frac{|z|^2 - |z| + 1}{2 + |z|} < \left(\frac{1}{\sqrt{3}} \right)^{-2}$$

$$\Rightarrow \frac{1}{3} < \frac{|z|^2 - |z| + 1}{2 + |z|} < 3$$

Let $|z| = t \geq 0$

$$\frac{1}{3} < \frac{t^2 - t + 1}{2 + t}$$

$$2 + t < 3t^2 - 3t + 3$$

$$\Rightarrow 3t^2 - 4t + 1 > 0$$

$$\Rightarrow (t-1)(3t-1) > 0$$

$$t < \frac{1}{3} \text{ or } t > 1 \quad \dots(i)$$

$$\frac{t^2 - t + 1}{2 + t} < 3$$

$$\Rightarrow t^2 - 4t - 5 < 0$$

$$\Rightarrow (t+1)(t-5) < 0$$

$$\Rightarrow -1 < t < 5 \quad \dots(ii)$$

From (i) and (ii)

$$t \in \left[0, \frac{1}{3} \right) \cup (1, 5) \Rightarrow 0 \leq |z| < \frac{1}{3} \text{ or } 1 < |z| < 5$$

3. Ans. (A,B,C,D)

Sol. Let $B \equiv (2t^2, 4t)$, then $C \equiv (2t^2, -4t)$

$$\text{then } -2 = 2 + (-t) \Rightarrow t = 1$$

$$B = (2, 4), C = (2, -4)$$

$$\text{or } ty = x + 2t^2$$

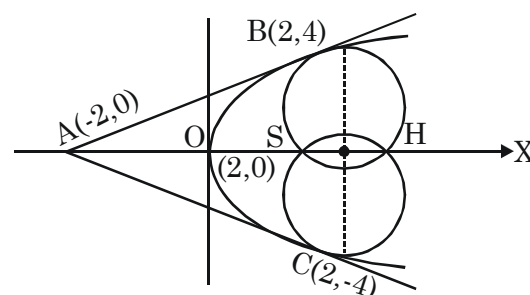
$$\text{Put } (-2, 0)$$

$$0 = -2 + 2t^2$$

$$t = \pm 1$$

$$\therefore (2t^2, 4t)$$

$$= (2, 4), (2, -4)$$



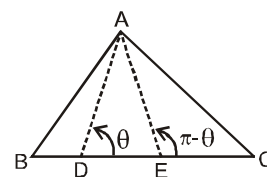
Clearly $AB = AC$, therefore AS is a median of $\triangle ABC$.

It is also the internal angular bisector $\angle BAC$ and it is also $\perp$ from A to opposite side BC . AS is also perpendicular bisector of side BC .

Hence, centroid, incentre, circumcentre and orthocentre lie on AS and hence also on common chord SH of the two circles.

4. Ans. (A,C, D)

Sol.



if we apply m-n Rule in $\triangle ABE$, we get

$$(1+1) \cot \theta = 1 \cdot \cot B - 1 \cdot \cot \theta$$

$$2 \cot \theta = \cot B - \cot \theta$$

$$3 \cot \theta = \cot B$$

$$\tan \theta = 3 \tan B \quad \dots(1)$$

Similarly, if we apply m-n Rule in $\triangle ACD$, we get

$$(1+1) \cot (\pi - \theta) = 1 \cdot \cot \theta - 1 \cdot \cot C$$

$$\cot C = 3 \cot \theta \Rightarrow \tan \theta = 3 \tan C \quad \dots(2)$$

form (1) and (2) we can say that

$$\tan B = \tan C \Rightarrow B = C$$

$$\therefore A + B + C = \pi$$

$$\therefore A = \pi - (B + C)$$

$$= \pi - 2B \quad \therefore B = C$$

$$\therefore \tan A = -\tan 2B$$

$$= -\left(\frac{2\tan B}{1-\tan^2 B}\right) = -\frac{\frac{2\tan \theta}{3}}{1-\frac{\tan^2 \theta}{9}}$$

$$\Rightarrow \tan A = \frac{6\tan \theta}{\tan^2 \theta - 9}$$

5. Ans. (A,B,D)

Sol. Characteristics equation for A :

$$\det(A - \lambda I_3) = 0$$

$$\begin{vmatrix} 1-\lambda & 2 & 2 \\ 2 & 1-\lambda & 2 \\ 2 & 2 & 1-\lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda^3 - 3\lambda^2 - 9\lambda - 5I = 0$$

Characteristics equation

$$\Rightarrow A^3 - 3A^2 - 9A - 5I = 0$$

To determine inverse :

$$(A^3 - 3A^2 - 9A - 5I)A^{-1} = 0$$

$$A^2 - 3A - 9I - 5A^{-1} = 0$$

$$\Rightarrow 5A^{-1} = A^2 - 3A - 9I$$

$$A^{-1} = \frac{1}{5} [A^2 - 3A - 9I]$$

$$A^2 = \begin{bmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{bmatrix}$$

[For the matrix to be invertible $\det A^2 \neq 0$]

$$\det A^2 \neq 0$$

$\Rightarrow A^2$ is invertible.

6. Ans. (A,C,D)

Sol. (A) Applying $\int_a^b f(a+b-x)dx$

$$I = \int_a^{\pi-a} (\pi-x)f(\sin(\pi-x))dx = \int_a^{\pi-a} (\pi-x)f(\sin x)dx$$

$$= \pi \int_a^{\pi-a} f(\sin x)dx - \int_a^{\pi-a} xf(\sin x)dx$$

$$I = \pi \int_a^{\pi-a} f(\sin x)dx - I$$

$$\Rightarrow 2I = \pi \int_a^{\pi-a} f(\sin x)dx$$

$$I = \frac{\pi}{2} \int_a^{\pi-a} f(\sin x)dx$$

(B) $\int_{-a}^a f^2(x)dx = 2 \int_0^a f^2(x)dx$ is true only when $f(x)$ is either even or odd.

(C) $\cos^2 x$ is periodic with fundamental period π

$$\int_0^{n\pi} f(\cos^2 x)dx = n \int_0^{\pi} f(\cos^2 x)dx$$

(D) Put $x+c=t \Rightarrow dx=dt$

$$x=0, t=c$$

$$x=b-c, t=b$$

$$\int_0^{b-c} f(x+c)dx = \int_c^b f(t)dt = \int_c^b f(x)dx$$

Paragraph for Question 7 and 8

7. Ans.(A)

8. Ans.(C)

Sol. $f(x) = x^3(x-2)^2(x-1)$

$$f'(x) = x^3(x-2)^2 + 2x^3(x-2)(x-1) + 3x^2(x-2)^2(x-1)$$

$$= x^2(x-2)(x^2-2x+2x^2-2x+3x^2-9x+6)$$

$$= x^2(x-2)(6x^2-13x+6)$$

$$f'(x) = x^2(x-2)(2x-3)(3x-2)$$

$$\Rightarrow \text{maxima at } x = \frac{3}{2} \text{ \& minima at } x = \frac{2}{3}, 2$$

$$\sum x_i^2 = \frac{9}{4} \text{ (For maxima)}$$

$$f(2) = 0$$

$$f\left(\frac{2}{3}\right) = \frac{8}{27} \times \frac{16}{9} \times \left(-\frac{1}{3}\right) = -\frac{128}{729}$$

9. Ans. (C)

10. Ans. (B)

Sol. Given function $f(x)$ can be rewritten as,

$$f(x) = \begin{cases} 0, & x < -1 \\ 1+x, & -1 \leq x \leq 0 \\ 1-x, & 0 < x \leq 1 \\ 0, & x > 1 \end{cases}$$

$$\Rightarrow f(x-1) = \begin{cases} 0, & x-1 < -1 \\ 1+(x-1), & -1 \leq x-1 \leq 0 \\ 1-(x-1), & 0 < x-1 \leq 1 \\ 0, & x-1 > 1 \end{cases}$$

$$\text{or } f(x-1) = \begin{cases} 0, & x < 0 \\ x, & 0 \leq x \leq 1 \\ 2-x, & 1 < x \leq 2 \\ 0, & x > 2 \end{cases}$$

$$\text{Also } f(x+1) = \begin{cases} 0, & x+1 < -1 \\ 1+(x+1), & -1 \leq x+1 \leq 0 \\ 1-(x+1), & 0 < x+1 \leq 1 \\ 0, & x+1 > 1 \end{cases}$$

$$\text{or } f(x+1) = \begin{cases} 0, & x < -2 \\ 2+x, & -2 \leq x < -1 \\ -x, & -1 < x \leq 0 \\ 0, & x > 0 \end{cases}$$

Now, $g(x) = f(x-1) + f(x+1)$

$$= \begin{cases} 0, & x < -2 \\ 2+x, & -2 < x \leq -1 \\ -x, & -1 < x \leq 0 \\ x, & 0 < x \leq 1 \\ 2-x, & 1 < x \leq 2 \\ 0, & x > 2 \end{cases}$$

It is easy to check that $g(x)$ is continuous for all $x \in \mathbb{R}$ and non-differentiable at $x = -2, -1, 0, 1, 2$

SECTION-II

1. Ans. (27.00)

Sol.: Let $O, A(\vec{a}), B(\vec{b}), C(\vec{c})$ be the vertices of given tetrahedron

$$V = \frac{1}{6} [\vec{a} \vec{b} \vec{c}]$$

The centroid are $\frac{\vec{a}+\vec{b}}{3}, \frac{\vec{b}+\vec{c}}{3}, \frac{\vec{c}+\vec{a}}{3}, \frac{\vec{a}+\vec{b}+\vec{c}}{3}$

$$V' = \frac{1}{6} \left[\frac{\vec{c}-\vec{a}}{3} \frac{\vec{c}-\vec{b}}{3} \frac{\vec{c}}{3} \right] = \frac{1}{6 \times 27} [\vec{a} \vec{b} \vec{c}] = \frac{1}{27} V$$

$$\therefore k = 27$$

2. Ans. (6.82 or 6.83)

Sol.: $\therefore$ Equation of circle

$$(x-2)^2 + (y+2)^2 + \lambda(x+y) = 0 \quad \dots(i)$$

$\therefore$ Centre lies on the x -axis

$\therefore \lambda = -4$ put in (i)

$\therefore$ equation of circle is $x^2 + y^2 - 8x + 8 = 0$

(α, β) lies on it

$$\Rightarrow \beta^2 = -\alpha^2 + 8\alpha - 8 \geq 0$$

$\therefore$ greatest value of ' α ' is $4 + 2\sqrt{2}$.

3. Ans. (13.66 or 13.67)

Sol.: $(2x^3 - y)dx + (x + 2yx^2)dy = 0$

$$2x dx + 2y dy + \frac{xdy - ydx}{x^2} = 0$$

$$\text{i.e. } d(x^2 + y^2) + d\left(\frac{y}{x}\right) = 0$$

$$\therefore d\left(x^2 + y^2 + \frac{y}{x}\right) = 0$$

$$\therefore p(x, y) = x^2 + y^2 + \frac{y}{x}$$

$$\text{Now } \frac{x-1}{2} = \frac{y+1}{3} = \frac{z+1}{1} = \lambda$$

$$\therefore x = 2\lambda + 1, y = 3\lambda - 1, z = \lambda - 1 = 0$$

$$\Rightarrow \lambda = 1$$

$$\therefore x = 3, y = 2$$

$$\therefore c = x^2 + y^2 + \frac{y}{x} = 9 + 4 + \frac{2}{3} = \frac{41}{3}$$

4. Ans. (7.00)

Sol.:

$$\begin{aligned}
 & \begin{array}{l} W_1 \\ W_2 \end{array} \begin{array}{l} \nearrow \\ \searrow \end{array} \begin{array}{l} \frac{1}{2} \xrightarrow{A} \frac{\frac{7!}{2!} + \frac{7!}{2!} - 6!}{8!} \\ \frac{1}{2} \xrightarrow{A} \frac{7!}{8!} = \frac{2 \times 7!}{8!} = \frac{1}{4} \end{array} \\
 & = \frac{4(7! - 6!)}{8!} = \frac{24 \cdot 6!}{8!} = \frac{3}{7} \\
 & \text{Required probability} = \frac{\frac{1}{2} \times \frac{3}{7}}{\frac{1}{2} \times \frac{3}{7} + \frac{1}{2} \times \frac{1}{4}} \\
 & = \frac{12}{19} = \frac{p}{q} \\
 & \therefore q - p = 7
 \end{aligned}$$

5. Ans. (80.00)

Sol.: $\vec{a}_1 = (a, b)$, $\vec{a}_2 = (a-2, b-\sqrt{5})$, $\vec{a}_3 = (a+2, b-\sqrt{5})$
Area of Δ , whose vertices are $\vec{a}_1, \vec{a}_2$ and $\vec{a}_3$, is

$$\begin{aligned}
 \Delta &= \frac{1}{2} \begin{vmatrix} 1 & a & b \\ 1 & a-2 & b-\sqrt{5} \\ 1 & a+2 & b-\sqrt{5} \end{vmatrix} \\
 \Rightarrow \Delta^2 &= \frac{1}{4} \begin{vmatrix} 1 & a & b \\ 1 & a-2 & b-\sqrt{5} \\ 1 & a+2 & b-\sqrt{5} \end{vmatrix} \begin{vmatrix} 1 & a & b \\ 1 & a-2 & b-\sqrt{5} \\ 1 & a+2 & b-\sqrt{5} \end{vmatrix} \text{ and}
 \end{aligned}$$

$$|\vec{a}_1 - \vec{a}_2| = 3, |\vec{a}_1 - \vec{a}_3| = 3, |\vec{a}_2 - \vec{a}_3| = 4$$

$$\Rightarrow \Delta = \sqrt{5 \times 2 \times 2 \times 1} = 2\sqrt{5}$$

$$(2\sqrt{5})^2 = \frac{1}{4} \begin{vmatrix} 1 + \vec{a}_1 \cdot \vec{a}_1 & 1 + \vec{a}_1 \cdot \vec{a}_2 & 1 + \vec{a}_1 \cdot \vec{a}_3 \\ 1 + \vec{a}_2 \cdot \vec{a}_1 & 1 + \vec{a}_2 \cdot \vec{a}_2 & 1 + \vec{a}_2 \cdot \vec{a}_3 \\ 1 + \vec{a}_3 \cdot \vec{a}_1 & 1 + \vec{a}_3 \cdot \vec{a}_2 & 1 + \vec{a}_3 \cdot \vec{a}_3 \end{vmatrix}$$

$\therefore$ Required answer = $4 \times 5 \times 4 = 80$

6. Ans. (2.00)

Sol.: $AB - BP = 2I$, $B = \text{adj}(B)$

$$\text{Let } B = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

$$\therefore \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$\therefore b = c = 0 \text{ and } a = d$$

$$\therefore B = \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix} = aI$$

$$aA - aP = 2I$$

$$aA = aP + 2I$$

$$\text{Let } P = \begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix} \text{ where } \begin{array}{l} \alpha\delta - \beta\gamma = 3 \\ \alpha + \delta = 4 \end{array}$$

$$\therefore aA = \begin{bmatrix} a\alpha + 2 & a\beta \\ a\gamma & a\delta + 2 \end{bmatrix}$$

$$\Rightarrow a^2 |A| = a^2(\alpha\delta - \beta\gamma) + 2a(\alpha + \delta) + 4$$

$$\Rightarrow a^2 |A| = 3a^2 + 8a + 4$$

$$|A| = 15 \quad T_r(B) < 0 \Rightarrow a < 0$$

$$\therefore 12a^2 - 8a - 4 = 0$$

$$a = -\frac{1}{3}, 1 \Rightarrow a = 1 (\because a < 0)$$

$$\therefore \text{Tr}(B) = 2a = -2/3$$

7. Ans. (16.00)

Sol.: $a_1 = 15$, $a_2 = 15 + d$, $a_3 = 15 + 2d$

$$a_{10} = 15 + 9d$$

$$\sum_{r=1}^{10} a_r^2 = 15^2 + (15 + d)^2 + \dots + (15 + 9d)^2 = 1185$$

$$15^2 \times 10 + d^2(1^2 + 2^2 + \dots + 9^2)$$

$$+ 30d(1 + 2 + \dots + 9) = 1185$$

$$2250 + d^2 \times \frac{9 \times 10 \times 19}{6} + 30d \times 45 = 1185$$

$$15 \times 19 d^2 + 30 \times 45d + 1065 = 0$$

$$19d^2 + 90d + 71 = 0$$

$$(19d + 71)(d + 1) = 0$$

$$\therefore d = -1, -\frac{71}{19} \text{ (as } a_2 \text{ is integer) (rejected)}$$

$$\therefore S(n) - S(n-1) \geq 0$$

$$\Rightarrow T_n \geq 0$$

$$\therefore n = 16$$

8. Ans. 7.00

$$S^2 = (1 + 3x + 5x^2 + \dots + 21x^{10})^2$$

$$(1 + 3x + 5x^2 + \dots + 21x^{10})$$

$$\text{Coeff. of } x^{18} = 21 \cdot 17 + 19 \cdot 19 + 17 \cdot 21 = 1075$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
FULL SYLLABUS
21-03-2021

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE-II)

PAPER-2

PART-1 : PHYSICS

SOLUTION

SECTION-I

1. Ans. (A,D)

Sol. $\pi r^2 \sqrt{2gy} = -\pi x^2 \frac{dy}{dx}$

$$r^2 \sqrt{2gy} dt = -\frac{y^{2/a}}{k^{2/a}} dy$$

$$k^{2/a} r^2 \sqrt{2g} \int_0^t dt = + \int_4^{16} y^{\frac{2}{a}-1} dy$$

(A) $k = 1, a = 4$

$$\sqrt{5} \times 10^{-4} \sqrt{20} t = 12$$

$$t = 12000 \text{ sec}$$

(B) $k = 1, a = 2$

$$r^2 \sqrt{2g} t = t \int_4^{16} y^{\frac{1}{2}} dy$$

$$r^2 \sqrt{2g} t = \frac{2^4 (16^{3/2} - 4^{3/2})}{3}$$

$$\sqrt{5} \times 10^{-4} \sqrt{20} t = \frac{2}{3} (56)$$

$$t = \frac{2(56) \times 10^4}{3\sqrt{5}\sqrt{20}}$$

$$= \frac{2}{3} (56) \times 1000$$

$$(C) (4) \sqrt{5} \times 10^{-4} \sqrt{20} t = \frac{2}{3} (56)$$

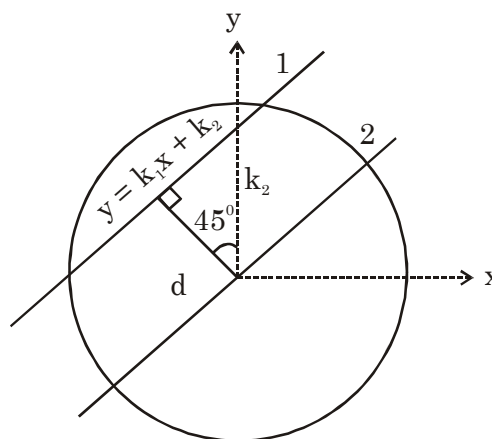
$$t = \frac{2(56) \times 1000}{3 \times 4}$$

$$(D) \sqrt{4}\sqrt{5} \times 10^{-4} \sqrt{20} t = \frac{2}{3} (56)$$

$$t = 6000 \text{ sec}$$

2. Ans. (B,C)

Sol.



$$I_1 = I_2 + md^2$$

$$I_1 = \frac{MR^2}{2} + M \left(\frac{k_2}{\sqrt{2}} \right)^2$$

$$(A) I_1 = \frac{MR^2}{2} + \frac{MR^2}{4} = \frac{3MR^2}{4}$$

$$(B) I_1 = \frac{MR^2}{2} + \frac{MR^2}{2} = MR^2$$

$$(C) I_1 = \frac{MR^2}{2} + \frac{MR^2}{2} = MR^2$$

$$(D) I_1 = \frac{MR^2}{2} + \frac{MR^2}{4} = \frac{3MR^2}{4}$$

3. Ans. (A,B,D)

Sol. Power leaving the system = P

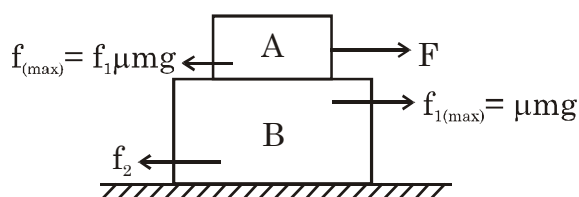
$$P = \sigma(4\pi C^2) T_C^4$$

Net power leaving B = P

$$P = \sigma(4\pi b^2) T_B^4 - \sigma(4\pi b^2) T_C^4$$

4. Ans. (A,B,C)

Sol.



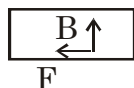
$$f_{(\max)} = 3\mu mg$$

Block 'B' always remains at rest as net force on B is zero.

$$F_{\text{GROUND}} = \sqrt{f^2 + \mu}$$

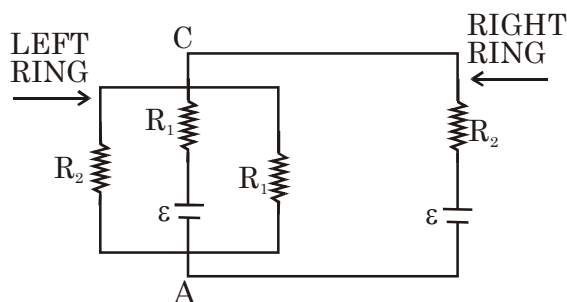
$$= \sqrt{(\mu mg)^2 + (3mg)^2}$$

$$= mg\sqrt{9 + \mu}$$



5. Ans. (A,D)

Sol. Simplified diagram may be



So there will be current in each ring and so magnetic force on them will be nonzero.

6. Ans. (A,D)

Sol. $30 = \frac{\varepsilon_1}{20}(4)$

$$\varepsilon_1 = 150$$

$$\varepsilon_2 = \frac{\varepsilon_1}{20} \times 6 = 45V$$

7. Ans. (B)

Sol. I $P \propto V$

$$w^+$$

$$\frac{I}{V} \propto V$$

$$T \propto V^2$$

$$U = +ve$$

$$Q = 4$$

II. w^+

$$\Delta U = 0$$

$$\theta = +$$

III $\Delta U = 0$

$$w^-$$

$$\theta = -$$

IV w^-

$$U = -V \propto T$$

$$\theta = -w$$

I A,C,D

II A,B,D

III B,E

IV B,E

8. Ans. (A)

Sol. (A) $\frac{1}{2}(P_0 + 2P_0)4V_0 = 6P_0V_0$

(B) AB $P \propto V$

$$T \propto V^2$$

$$\frac{T_A}{T_B} = \left(\frac{4V_0}{8V_0}\right)^2$$

$$T_B = 400 \text{ K.}$$

$$\Delta U_{AB} = nCUDT$$

$$-\frac{3}{25} \frac{3}{2} \left(\frac{25}{3}\right) \times 300 = 450J$$

(C) WCYCLIZ = _____

$$-(4P_0)7V_0 + \frac{1}{2}(R_0 + 4R_0)3V_0 + P_0V_0$$

$$-\left(28 + \frac{15}{2} + 1\right)P_0V_0 = \frac{-39}{2}P_0V_0$$

(D) $UCD = U_{CD} = nCv\Delta T = \frac{3}{25} \frac{3}{2}(-700)$

$$T_B = 400 \text{ L}$$

$$T_C = 800 \text{ K}$$

$$T_B = 100 \text{ K}$$

$$CD = V \propto T$$

9. Ans. (A)

Sol. $F = \frac{mv^2}{r} = \frac{ke^2}{r^4} \dots (1) \quad v^2 \propto \frac{1}{r^3}$

$$mvr = \frac{nh}{2\pi} \dots (2) \quad vr \propto n$$

$$v^2 \propto \frac{v^3}{n^3} \quad \boxed{v \propto n^3}$$

$$r \propto n^{-2}$$

$$(A) I = \frac{dq}{dt} = \frac{e}{T} = \frac{e}{2\pi r} v$$

$$I \propto \frac{v}{r} \propto \frac{n^3}{n^{-2}}$$

$$(B) B = \frac{\mu_0 I}{2\pi r} \propto \frac{n^5}{n^{-2}}$$

$$(C) L = mvr \propto n$$

$$(D) \mu = NIA \propto n^5 n^{-4}$$

$$\mu \propto n$$

10. Ans. (B)

Sol. $U = U_0 \ln\left(\frac{r}{r_0}\right)$

$$F = \frac{mv^2}{r} = \frac{U_0 r_0}{r} \frac{1}{r_0} = \frac{U_0}{r}$$

$$v \propto r^0$$

$$mvr = \frac{nh}{2\pi}$$

$$vr \propto n$$

$$r \propto n$$

$$v \propto n^0$$

$$(A) I \propto \frac{v}{r} \propto n^{-1}$$

$$(B) B \propto \frac{I}{r} \propto \frac{v}{r^2} \propto n^{-2}$$

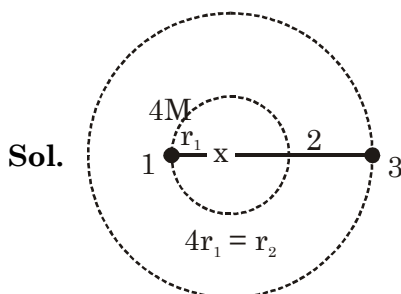
$$(C) L = mvr \propto n$$

$$(D) \mu = NIA \propto \frac{v}{r} r^2 \propto vr$$

$$\mu \propto n$$

SECTION-II

1. Ans. 0.25

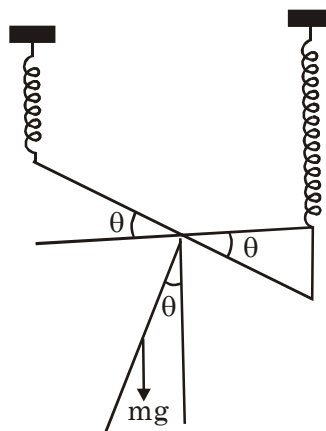


$$\frac{K.E_{4m}}{K.E_m} = \frac{\frac{1}{2} 4mv_1^2}{mv_2^2}$$

$$= \frac{4r_1^2 \omega^2}{r_2^2 \omega^2} = \frac{4r_1^2}{r_2^2} = \frac{1}{4}$$

2. Ans. 12.00

Sol. Let the body be displaced through an angle θ about its mean position.



Net torque on the body at this position about C.

$$\tau = -(2Ka\theta a \cos \theta + mga \sin \theta)$$

$$= -(2Ka^2 + mga)\theta$$

(where θ is small, $\sin \theta = \theta$ and $\cos \theta = 1$)

Angular acceleration

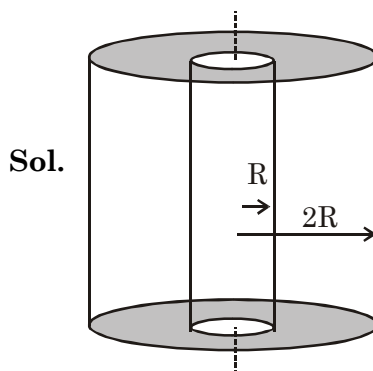
$$\alpha = \frac{\tau}{I} = -\frac{(2Ka^2 + mga)\theta}{\frac{m \times (2a)^2}{12} + \frac{m \times (2a)^2}{3}}$$

$$\text{or } \alpha = -\frac{(2Ka^2 + mga)}{\frac{5ma^2}{3}}$$

$$\Rightarrow \omega = \sqrt{\frac{(2Ka^2 + mga)}{5ma^2/3}} = \sqrt{\left(\frac{6K}{5m} + \frac{3g}{5a}\right)}$$

$$= \sqrt{\frac{6}{5} \times \frac{96}{6} + \frac{30}{5 \times 5}} \times 10^4 = 12 \text{ rad/s}$$

3. Ans. 6.00



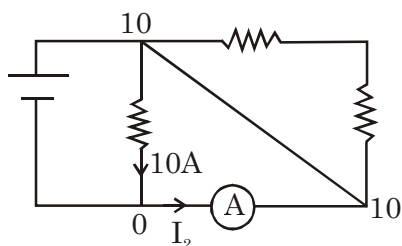
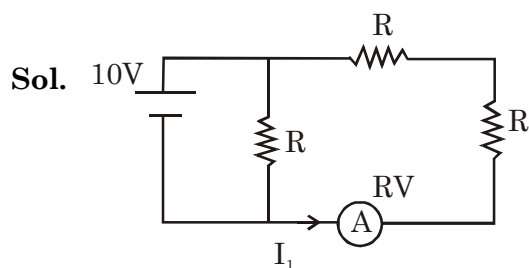
$$H = -K, 2\pi r l \frac{dT}{dr}$$

$$\int_{R_1}^{R_2} \frac{H dr}{2\pi r l} = -K \int_{T_1}^{T_2} dT$$

$$H = \frac{2\pi / k (T_1 - T_2)}{\ln \frac{R_2}{R_1}} H_i = H_f$$

$$\therefore n = 6$$

4. Ans. 1.33



$$I_2 = 4I_1$$

$$\frac{10}{RV} = \frac{4(10)}{(4 + RV)}$$

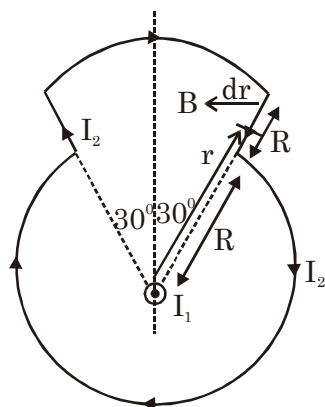
$$4 + RV = 4RV$$

$$3RV = 4$$

$$RV = 1.33$$

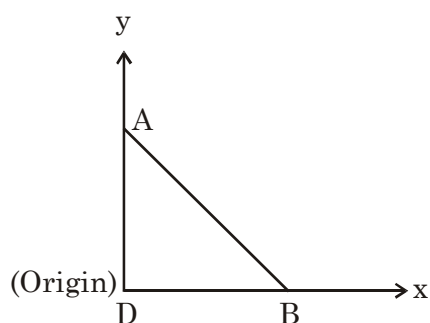
5. Ans. 2.00

Sol. $\int d\tau = 2 \int_R^{2R} \frac{\mu_0 I_1 I_2}{2\pi x} \times \sin 30^\circ dx$
 $\tau = 20$



6. Ans. 6.66 to 6.67

Sol.



Potential on vertex of cube of side a is V_0

$$\text{Let } V \propto \frac{1}{4\pi\epsilon_0} \rho^x a^y$$

Dimensionally $x = 1, y = 2$

If side is made half potential at vertex
 $= \frac{V_0}{4}$

Net potential at centre due to eight cubes

$$\text{of side } \frac{a}{2} = 8 \times \frac{V_0}{4}$$

$$8 \times \frac{V_0}{4} = 2V_0$$

7. Ans. 3.00

Sol. $\ell = 16.2 \pm 0.1 \text{ cm}$

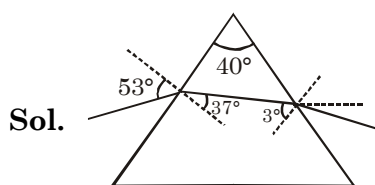
$$b = 10.1 \pm 0.1 \text{ cm}$$

$$\ell b = 163.62 \pm 2.6 \text{ cm}^2$$

using proper rules

$$\ell b = 164 \pm 3 \text{ cm}^2$$

8. Ans. 17.00



$$1 \sin 53 = \frac{4}{3} \sin r_1$$

$$\frac{4}{5} = \frac{4}{3} \sin r_1$$

$$\Rightarrow r_1 = 37^\circ \Rightarrow r_2 = 3^\circ$$

$$\frac{4}{3} \times \sin 3^\circ = \sin e \Rightarrow e = 4^\circ$$

$$\delta = i + e - A$$

$$= 53 + 4 - 40$$

$$= 17^\circ$$

PART-2 : CHEMISTRY

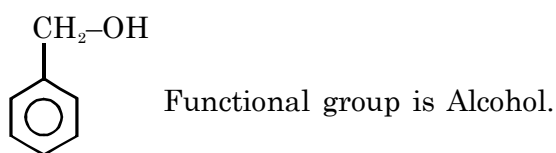
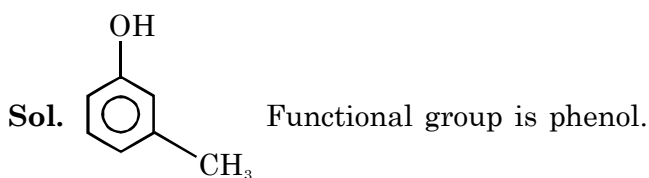
SOLUTION

SECTION-I

1. Ans. (B,C)

2. Ans. (B,C)

3. Ans. (B)



4. Ans. (A,C)

Sol. $X^+ + H_2O \rightleftharpoons XOH + H^+$

$$K_H = \frac{k_w}{k_b} = \frac{10^{-14}}{4 \times 10^{-5}} = 2.5 \times 10^{-10}$$

$$CH^2 \approx k_H$$

$$(H^+) = Ch = \sqrt{c \cdot k_H}$$

$$10^{-5} = \sqrt{c \times 2.5 \times 10^{-10}}$$

$$C = 0.4 \text{ molar}$$

$$0.4 = \frac{80}{M_{xy}} \times \frac{1}{2} \Rightarrow M_{xy} = \frac{40}{0.4} = 100$$

$$h = \sqrt{\frac{k_H}{C}} = \sqrt{\frac{2.5 \times 10^{-10}}{0.4}}$$

$$h = 2.5 \times 10^{-5}$$

$$\%h = 2.5 \times 10^{-3}$$

$$r_{x^+} + r_{y^-} = \sqrt{3} \frac{a}{2}$$

$$1.6 + 1.864 = r_{x^+} + r_{y^-} = \frac{\sqrt{3}}{2} a$$

$$a = 4\text{\AA}$$

$$\text{density} = \frac{1 \times \frac{100}{N_0}}{(4 \times 10^{-8})^3} \text{ gm/cc}$$

$$= 2.6 \text{ gm/cc.}$$

5. Ans. (A,B,D)

6. Ans. (A,B,C,D)

7. Ans. (B)

Sol. P-(3); Q-(1),(5); R-(1),(2),(5); S-(3)

8. Ans. (C)

9. Ans. (B)

10. Ans. (C)

Sol.

$$(P) \quad E_{\text{cell}} = -\frac{0.0591}{2} \log \frac{C}{2C} = 8.89 \times 10^{-3}$$

Spontaneous, cell works, conc. cell.

$$(Q) \quad E_{\text{cell}} = -0.0591 \log \frac{1}{1} = 0$$

Non spontaneous, cell will not work conc. cell.

$$(R) \quad E_{\text{cell}} = -\frac{0.0591}{2} \log \frac{0.01}{0.1^2} - 0.74 + 0.8 = 0.06$$

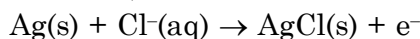
Spontaneous, cell works.

$$(S) \quad E_{\text{cell}} = -\frac{0.0591}{1} \log \frac{Cl^-}{K_{sp}} - \frac{0.0591}{1} \log \frac{1}{[Ag^+]}$$

$$= \frac{0.0591}{2} \log \frac{0.1 \times 0.1}{K_{sp}(AgCl)} = +ve$$

Spontaneous, cell works, conc. cell.

at anode,



SECTION-II

1. Ans. (5.66)

Sol.: [Range 5.5 to 5.7]

$$PI = \frac{2.2 + 9.1}{2} = \frac{11.3}{2} = 5.66$$

2. Ans. (6.00)

Sol.: 2, 3, 4, 6, 7, 8

With coordination number six and oxidation state +2, wo (n - 1)d orbitals are not available for d^2sp^3 .

3. Ans. (9.10 to 9.40)

$$\text{Sol.: } z = 1 + \frac{Pb}{RT}$$

$$\frac{dz}{dp} = \frac{1}{10} = \frac{L}{RT}$$

$$b = \frac{RT}{10} = \frac{22.4}{10} = 2.24$$

$$4 \times V \times N_0 = 2.24$$

$$V = \frac{2.24}{4 \times 6} \times 10^{-23} \ell$$

$$= \frac{2.24 \times 1000}{24} \times 10^{-23} \text{ cm}^3$$

$$= 9.33 \times 10^{-22} \text{ cm}^3$$

4. Ans. (36.00)

Sol.: (a) ZrI_4 Oxidation state = 4

(b) $[\text{Ni}(\text{CO})_4]$ Bond order = 3

(c) $\text{Na}_2 [\text{Zn}(\text{CN})_4] = 3$

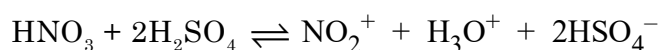
5. Ans. (6.00)

Sol. All aldehyde hemeacetal α -hydroxy keto group, HCHO , can be oxidised by Tollen's reagent.

(i), (ii), (iii), (v), (vi), (viii)

6. Ans. (0.18 to 0.20)

Sol.:



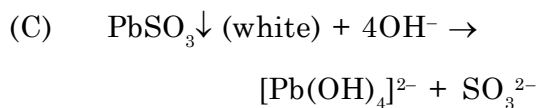
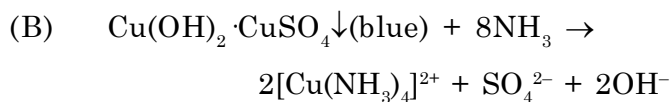
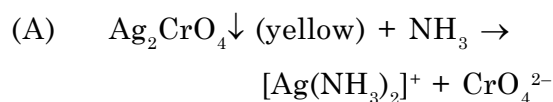
$$\begin{array}{ccccc} 1 & 100 & & & \\ 0.7 & 99.7 & 0.3 & 0.3 & 0.6 \\ \cong & 100 & & & \end{array}$$

$$\text{Total moles of particles} = 0.7 + 0.3 + 0.3 + 0.6 = 1.9$$

$$\therefore m = 1.9 \times \frac{1000}{9800} \approx 0.19$$

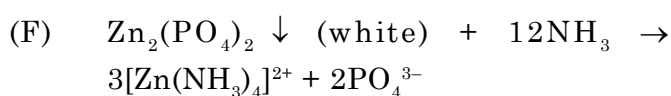
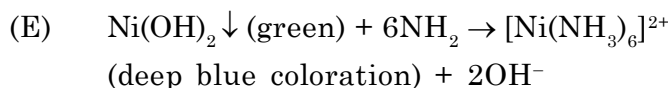
7. Ans. (4.00)

Sol.:



Soluble in sodium hydroxide not in NH_3 .

(D) Soluble in sodium hydroxide not in NH_3 .



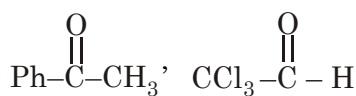
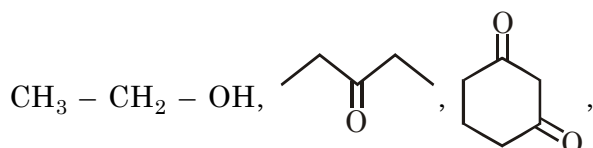
(G) Insoluble in NH_3 , soluble appreciably in boiling concentrated H_2SO_4 .

(H) $\text{Bi}(\text{OH})_3$ is insoluble in NH_3 .

(I) $\text{Mn}(\text{OH})_2$ is insoluble in excess conc. solution of ammonia

8. Ans. (5.00)

Sol. Chloroform given by



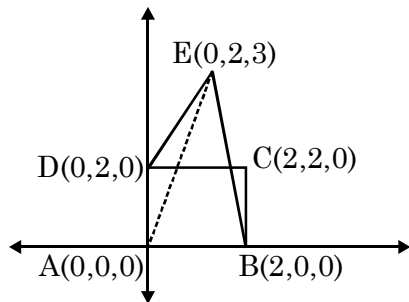
PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. Ans. (A,C)

Sol. Plane ABE is $\begin{vmatrix} x & y & z \\ 0 & 2 & 3 \\ 2 & 0 & 0 \end{vmatrix} = 0$



$$x(0) - y(-6) + z(-4) = 0$$

$$\Rightarrow 3y - 2z = 0$$

2. Ans. (A,C,D)

Sol. z_1, z_2 lie on the circle whose centre is $(-2, -3)$ and radius is equal to.

$$\therefore \arg(z_3 - z_1) = \arg(z_3 - z_2)$$

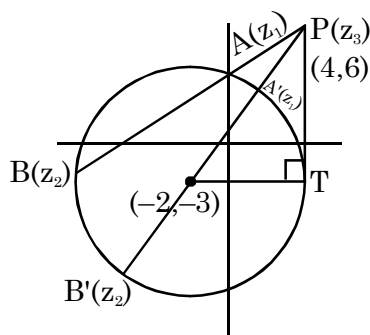
$\Rightarrow z_1, z_2$ and z_3 are collinear

$$PA \cdot PB = PT^2$$

$$|z_3 - z_1| |z_3 - z_2| = OP^2 - OT^2$$

$$= 36 + 81 - 16$$

$$= 101$$



$|z_1 - z_2|$ is maximum $\Rightarrow z_1$ and z_2 are extremities of diameter, $|z_1| = 4 - \sqrt{13}$

and $z_2 = 4 + \sqrt{13}$

$$\therefore |z_1 z_2| = 16 - 13 = 3$$

$$|z_1| + |z_2| = 8$$

3. Ans. (B,C)

Sol. $(A^{-1}BA)^{-1} = 2(\text{adj } A)$

$$A^{-1}BA = 2 \text{ adj } A$$

$$BA = 2A (\text{adj } A) = 2|A| I_3$$

$$B = 2|A| A^{-1} = 10A^{-1}$$

$$\therefore \det(B) = 10^3 \times \frac{1}{5} = 200$$

$$\text{Tr.}(B) = 10 \text{ Tr.}(A^{-1}) = 10 \times 3 = 30$$

4. Ans. (B,D)

Sol. As tangents are perpendicular to each other and its points of intersections

$P\left(\frac{7}{5}, -\frac{4}{5}\right)$ lies on directrix. AB is focal chord.

Equation of AB is $x - y = 4$, as focus is (α, β)

$$\text{so } \alpha - \beta = 4 \Rightarrow \beta = \alpha - 4$$

Now $\angle ASP = 90^\circ$

$$\Rightarrow \frac{-2 - \alpha + 4}{2 - \alpha} \cdot \frac{-\frac{4}{5} - \alpha + 4}{\frac{7}{5} - \alpha}$$

$$\Rightarrow \frac{2 - \alpha}{2 - \alpha} \cdot \frac{\frac{16}{5} - \alpha}{\frac{7}{5} - \alpha} = -1$$

$$\Rightarrow \frac{16}{5} - \alpha = \alpha - \frac{7}{5}; \alpha = \frac{23}{10}$$

$$\Rightarrow \text{Focus is } \left(\frac{23}{10}, -\frac{17}{10}\right)$$

$$\text{Also, } \ell_1 = AS = \frac{3\sqrt{2}}{10}, \ell_2 = BS = \frac{27\sqrt{2}}{10}$$

$$\frac{1}{a} = \frac{1}{\ell_1} + \frac{1}{\ell_2} \Rightarrow a = \frac{27\sqrt{2}}{100} \Rightarrow 4a = \frac{27\sqrt{2}}{25}$$

5. Ans. (A,C)

Sol. $I_0 = \int_0^{\pi/4} \frac{\left(x - \frac{\pi}{4}\right)^{50}}{I} \cdot \frac{\sec^2 x \, dx}{II}$

$$I_0 = \left[\left(x - \frac{\pi}{4}\right)^{50} \cdot \tan x \right]_0^{\pi/4}$$

$$- 50 \int_0^{\pi/4} \left(x - \frac{\pi}{4}\right)^{49} \cdot \tan x \, dx$$

$$\therefore I_0 = -50 \int_0^{\pi/4} \left(x - \frac{\pi}{4}\right)^{49} \tan x \, dx$$

$$\therefore I_1 = \frac{\int_0^{\pi/4} \left(x - \frac{\pi}{4}\right)^{50} \sec^2 x \, dx}{\int_0^{\pi/4} \left(x - \frac{\pi}{4}\right)^{49} \cdot \tan x \, dx} = -50$$

$$I_2 = \frac{1}{100} \int_0^{\pi/4} \frac{\left(x - \frac{\pi}{4}\right)^{100}}{I} \cdot \frac{\sec^2 x \, dx}{II}$$

$$- \int_0^{\pi/4} x^{99} \left(\frac{1 - \tan x}{1 + \tan x} \right) dx$$

Apply King prop.,

$$I_2 = 0$$

$$\therefore I_1 < I_2$$

$$I_1 \cdot I_2 = 0$$

6. Ans. (B,C)

Sol. $\frac{\cos A}{a} = \frac{b^2 + c^2 - a^2}{2abc}, \frac{\cos B}{b} = \frac{c^2 + a^2 - b^2}{2abc}, \frac{\cos C}{c} = \frac{a^2 + b^2 - c^2}{2abc}$

$$\therefore \frac{\cos A}{a} + \frac{\cos B}{b} + \frac{\cos C}{c} = \frac{a^2 + b^2 + c^2}{2abc}$$

$$\& \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} = \frac{s-a}{\Delta} + \frac{s-b}{\Delta} + \frac{s-c}{\Delta}$$

$$= \frac{3s-2s}{\Delta} = \frac{s}{\Delta} = \frac{1}{r}$$

7. Ans. (A)

Sol.

$$\frac{d(g(x))}{d f(x)} = \begin{cases} \frac{-2}{1 + (f(x))^2}, & f(x) \in (-1, 1) \\ \frac{-2}{1 + (f(x))^2}, & f(x) \in (-\infty, -1) \\ \frac{-2}{1 + (f(x))^2}, & f(x) \in (1, \infty) \end{cases}$$

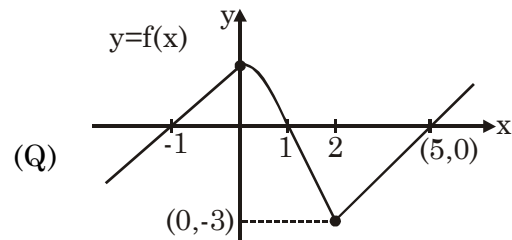
(P) $\frac{-2}{1 + (f(x))^2} = -\frac{1}{13}$

$$\Rightarrow 1 + (f(x))^2 = 26$$

$$\Rightarrow f(x) = \pm 5$$

$$f(x) = 5 \text{ when } x = 10$$

$$f(x) = -5 \text{ when } x = -6$$



$f(x)$ has local maxima at $x = 0$ and local minima at $x = 2$.

(R) From the graph of $y = f(x)$ we can see that $f(x) = -k$ has 2 positive and one negative root if $-k \in (-3, 1)$

$$\Rightarrow k \in (-1, 3)$$

(S) $g'(x) = \frac{-2}{1 + (f(x))^2} \times f'(x)$

$$g'(x) < 0 \Rightarrow f'(x) > 0$$

$$\Rightarrow x \in (-\infty, 0) \cup (2, \infty)$$

8. Ans. (C)

Sol.

(P) $\frac{d^2 y}{dx^2} = 6x - 4$

$$\Rightarrow \frac{dy}{dx} = 3x^2 - 4x + c$$

$$\text{at } x = 1, \frac{dy}{dx} = 0 \Rightarrow c = 1$$

$$\therefore \frac{dy}{dx} = 3x^2 - 4x + 1$$

$$\Rightarrow y = x^3 - 2x^2 + x + d$$

$$\text{at } x = 1, y = 5$$

$$\Rightarrow 5 = 1 - 2 + 1 + d \Rightarrow d = 5$$

$$\therefore f(x) = x^3 - 2x^2 + x + 5$$

$$f(0) = 5$$

$$(Q) \quad \frac{dy}{dx} = 2ax + b$$

$$\left. \frac{dy}{dx} \right|_{(1,2)} = 2a + b$$

$$y = ax^2 + bx + \frac{7}{2} \text{ passes through } (1, 2)$$

$$\Rightarrow 2 = a + b + \frac{7}{2}$$

$$\Rightarrow \boxed{a + b = \frac{-3}{2}} \quad \dots(1)$$

$$y = x^2 + 6x + 10$$

$$\frac{dy}{dx} = 2x + 6$$

$$\text{Slope of normal at } (-2, 2) = \frac{-1}{2(-2) + 6}$$

$$= -\frac{1}{2}$$

$$\therefore \boxed{2a + b = -\frac{1}{2}} \quad \dots(2)$$

From (1) and (2)

$$a = 1, b = \frac{-5}{2}$$

$$\therefore \frac{a}{2} - b = \frac{1}{2} + \frac{5}{2} = 3$$

$$(R) \quad 5\alpha(3x^2) + 10\alpha(2x) + 1 + 2\frac{dy}{dx} = 0$$

$$\Rightarrow \frac{dy}{dx} = \frac{-15\alpha^2 x^2 - 20\alpha x - 1}{2}$$

$$\Rightarrow \frac{dy}{dx} \text{ at } A = \frac{-1}{2} \quad A : (0, 2)$$

Equation of normal at A is

$$y - 2 = 2(x - 0)$$

$$\Rightarrow y = 2x + 2$$

Let normal meets the curve at B.

$$\Rightarrow 5\alpha^2 x^3 + 10\alpha x^2 + x + 4x + 4 - 4 = 0$$

$$\Rightarrow 5x(\alpha x + 1)^2 = 0$$

$$\Rightarrow x = \frac{-1}{\alpha}$$

$$\therefore B : \left(-\frac{1}{\alpha}, \frac{-2}{\alpha} + 2 \right)$$

Slope of tangent at B

$$= \frac{-\frac{15\alpha^2}{\alpha^2} + \frac{20\alpha}{\alpha} - 1}{2} = 2$$

$\therefore$ Equation of tangent is

$$y + \frac{2}{\alpha} - 2 = 2 \left(x + \frac{1}{\alpha} \right)$$

$$\Rightarrow \boxed{y = 2x + 2}$$

$$(S) \quad f(x) = \frac{px}{e^x} - \frac{x^2}{2} + x$$

$$f'(x) = -px e^{-x} + pe^{-x} - x + 1$$

$$f'(x) = pe^{-x}(-x + 1) + (-x + 1)$$

$$= (pe^{-x} + 1)(1 - x) \leq 0, \quad \forall x < 0$$

$$\Rightarrow p \leq -1$$

9. Ans. (A)

Sol.

$$(P) \quad f(x) = \text{sgn}(\sin^2 x - \sin x - 1)$$

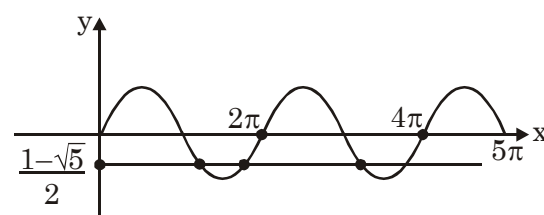
$$\sin^2 x - \sin x - 1 = 0$$

$$\Rightarrow \sin x = \frac{1 \pm \sqrt{5}}{2}$$

$$\Rightarrow \sin x = \frac{1 - \sqrt{5}}{2}$$

$f(x)$ will be discontinuous when

$$\sin x = \frac{1 - \sqrt{5}}{2}$$



$$\Rightarrow x \in (0, 4\pi] \text{ or } x \in (0, 5\pi]$$

(Q) $y = \left[\frac{f(x)}{g(x)} \right]$

$$y = \left[\frac{x^2 - x + k - 2}{x^2 - x + 1} \right]$$

$$y = \left[1 + \frac{k-3}{x^2 - x + 1} \right]$$

$$y = 1 + \left[\frac{k-3}{\left(x - \frac{1}{2}\right)^2 + \frac{3}{4}} \right]$$

for y to be continuous in R

$$0 \leq k-3 < \frac{3}{4} \Rightarrow k \in \left[3, \frac{15}{4} \right)$$

$$\Rightarrow a = 3, b = \frac{15}{4}$$

$$b - \frac{5a}{4} = \frac{15}{4} - \frac{15}{4} = 0$$

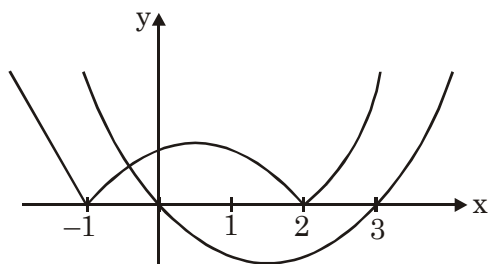
(R) $f(x) = \begin{cases} \cos^{-1} x, & \frac{1}{2} \leq |x| \leq 1 \\ \frac{2|x|}{3} + \frac{\pi}{3} - \frac{1}{3}, & |x| < \frac{1}{2} \end{cases}$

$$f(x) = \begin{cases} \cos^{-1} x, & x \in \left[-1, -\frac{1}{2}\right] \cup \left[\frac{1}{2}, 1\right] \\ -\frac{2x}{3} + \frac{\pi}{3} - \frac{1}{3}, & x \in \left(-\frac{1}{2}, 0\right) \\ \frac{2x}{3} + \frac{\pi}{3} - \frac{1}{3}, & x \in \left(0, \frac{1}{2}\right) \end{cases}$$

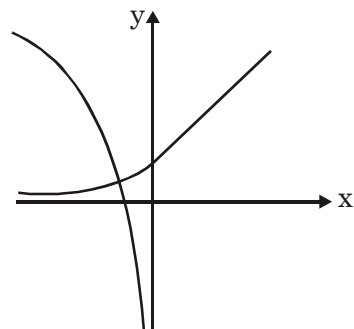
$\cos^{-1} x$ is non differentiable at $x = -1, 1$.

$f(x)$ is non-differentiable at $x = -1, -\frac{1}{2}, 0, \frac{1}{2}, 1$.

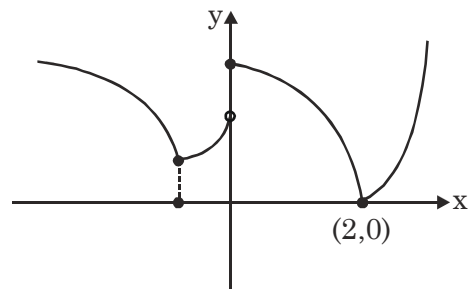
(S) $\max [|x^2 - x - 2|, x^2 - 3x]$



$$\max (\ln(-x), e^x)$$



$$y = f(x)$$



From the above graph of $y = f(x)$ we can see that $f(x)$ is non differentiable at 3 points.

10. Ans. (A)

Sol.

(P) If $a = 1$

$$f(x) = \lim_{t \rightarrow \infty} \frac{(1 + \sin \pi x)^t - 1}{(1 + \sin \pi x)^t + 1}$$

If $x \in (0, 1) \cup (2, 3) \cup (4, 5)$

$$f(x) = 1$$

If $x \in (1, 2) \cup (3, 4) \cup (5, 6)$

$$f(x) = -1$$

If $x = 2, 3, 4, 5$

$$f(x) = 0$$

$\therefore f(x)$ is discontinuous at $x = 1, 2, 3, 4, 5$

(Q) If $a = 3$

$$f(x) = \lim_{t \rightarrow \infty} \frac{(3 + \sin \pi x)^t - 1}{(3 + \sin \pi x)^t + 1}$$

$$f(x) = 1 \quad \forall x \in \mathbb{R}$$

$\therefore f(x)$ is continuous $\forall x \in \mathbb{R}$

(R) If $a = 0.5$

$$f(x) = \lim_{x \rightarrow \infty} \frac{(0.5 + \sin \pi x)^t - 1}{(0.5 + \sin \pi x)^t + 1}$$

Limit does not exist at

$$x = \frac{1}{6}, \frac{5}{6}, \frac{13}{6}, \frac{17}{6}, \frac{25}{6}, \frac{29}{6}$$

(S) If $a = 0$

$$f(x) = \lim_{t \rightarrow \infty} \frac{(\sin \pi x)^t - 1}{(\sin \pi x)^t + 1}$$

(i) If $x = \frac{1}{2}, \frac{5}{2}, \frac{9}{2}$

$$f(x) = 0$$

(ii) If $x \in \left(0, \frac{1}{2}\right) \cup \left(\frac{1}{2}, \frac{3}{2}\right) \cup \left(\frac{3}{2}, \frac{5}{2}\right) \cup \left(\frac{5}{2}, \frac{7}{2}\right)$

$$\cup \left(\frac{7}{2}, \frac{9}{2}\right) \cup \left(\frac{9}{2}, \frac{11}{2}\right) \cup \left(\frac{11}{2}, 6\right)$$

$$f(x) = -1$$

(iii) If $x = \frac{3}{2}, \frac{7}{2}, \frac{11}{2}$

$f(x)$ is not defined

$\therefore f(x)$ is discontinuous at

$$x = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \frac{7}{2}, \frac{9}{2}, \frac{11}{2}$$

SECTION-II

1. Ans. (1.00)

Sol.: $(\vec{r} - \vec{a}) \times (\vec{b} - \vec{a})$ is parallel to $\hat{n}$

$$\Rightarrow (\vec{r} - \vec{a}) \times (\vec{b} - \vec{a}) = t \hat{n} \quad \dots(1)$$

$$\therefore |(\vec{r} - \vec{a}) \times (\vec{b} - \vec{a}) \cdot \hat{n}| = t$$

$$|[\vec{r} - \vec{a} \quad \vec{b} - \vec{a} \quad \hat{n}]| = |t| \quad \dots(2)$$

From (1)

$$\vec{r} \times (\vec{b} - \vec{a}) - \vec{a} \times \vec{b} = t \hat{n}$$

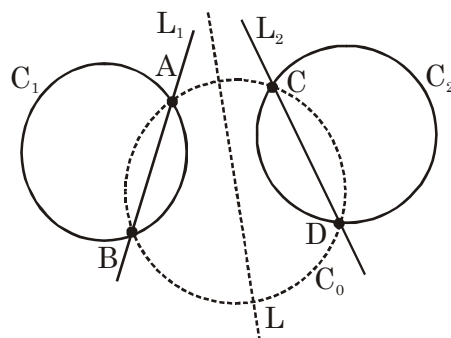
$$\text{or } |\vec{r} \times (\vec{b} - \vec{a}) - (\vec{a} \times \vec{b})| = |t| \quad \dots(3)$$

From (2) and (3)

$$|[\vec{r} - \vec{a} \quad \vec{b} - \vec{a} \quad \hat{n}]| = |\vec{r} \times (\vec{b} - \vec{a}) - (\vec{a} \times \vec{b})|$$

2. Ans. (4.00)

Sol.: We represent the given situation through the following diagram :



• C_0 is the circle passing through A, B, C and D.

• L is the radical axis of C_1 and C_2 given by.

$$S_1 - S_2 = 0$$

$$\Rightarrow 2x - 2y - 6 = 0$$

We use the fact that the radical axes of three circles taken two at a time are concurrent. Thus, L , L_1 and L_2 are concurrent :

$$\begin{vmatrix} 2 & -2 & -6 \\ 1 & 2 & 3 \\ 2 & 3 & \lambda \end{vmatrix} = 0 \Rightarrow \lambda = 4$$

3. Ans. (1.00)

Sol. $\sin y \frac{dy}{dx} - 2 \cos y \cos x = -\sin^2 x \cos x$

Put $-\cos y = z$

$$\Rightarrow \sin y \frac{dy}{dx} = \frac{dz}{dx}$$

$$\frac{dz}{dx} + (2 \cos x)z = -\sin^2 x \cos x$$

$$\text{I.F.} = e^{\int 2 \cos x dx} = e^{2 \sin x}$$

$$z \cdot e^{2 \sin x} = -\int e^{2 \sin x} \cos x \sin^2 x dx$$

$$\therefore z = -\cos y = \frac{-\sin^2 x}{2} - \frac{1}{4} + \frac{\sin x}{2} + ce^{-2 \sin x}$$

$y = 0$ at $x = 0$

$$\Rightarrow -1 = 0 - \frac{1}{4} + 0 + c$$

$$\Rightarrow c = -\frac{3}{4}$$

$$\therefore \cos y = \frac{1}{2} \sin^2 x - \frac{1}{2} \sin x + \frac{1}{4} + \frac{3}{4} e^{-2 \sin x}$$

$$\therefore A + B + C + D = 1$$

4. **Ans. (6.00)**

Sol.: Let $P(E_k) = \lambda K(K+1)$

$$\sum_{k=1}^n P(E_k) = 1$$

$$\Rightarrow \lambda \sum_{k=1}^n k(k+1) = 1$$

$$\Rightarrow \lambda = \frac{3}{n(n+1)(n+2)}$$

Let $P(E)$ denote the probability that a coin selected at random is biased.

$$P(E_1/E) = \frac{P(E_1) \times P(E/E_1)}{\sum_{k=1}^n P(E_k) \times P(E/E_k)}$$

$$P(E_1/E) = \frac{2\lambda \times \frac{1}{n}}{\sum_{k=1}^n \lambda \cdot k \cdot (k+1) \cdot \frac{k}{n}}$$

$$= \frac{2}{\sum_{k=1}^n k^2(k+1)}$$

$$= \frac{24}{n(n+1)(n+2)(3n+1)}$$

$$\therefore F(n) = \frac{24}{n(n+1)(n+2)(3n+1)}$$

$$\sum_{r=1}^k F(r) \cdot (3r+1)$$

$$= \sum_{r=1}^k \frac{24}{r(r+1)(r+2)}$$

$$= 12 \sum_{r=1}^k \left(\frac{1}{r(r+1)} - \frac{1}{(r+1)(r+2)} \right)$$

$$= 12 \left(\frac{1}{1 \cdot 2} - \frac{1}{(k+1)(k+2)} \right)$$

$$\therefore \lim_{k \rightarrow \infty} \sum_{r=1}^k F(r) (3r+1) = \frac{12}{2} = 6$$

5. **Ans. (8.00)**

Sol.: $\frac{1}{a}, \frac{1}{a_2}, \frac{1}{a_3}, \dots, \frac{1}{a_9}$ are in A.P.

$$d = \frac{1}{4} - \frac{1}{5} = \frac{1}{20}$$

Series is $\frac{1}{20}, \frac{2}{20}, \frac{3}{20}, \dots, \frac{9}{20}$

$$D = \begin{vmatrix} 20 & \frac{20}{2} & \frac{20}{3} \\ \frac{20}{4} & \frac{20}{5} & \frac{20}{6} \\ \frac{20}{7} & \frac{20}{8} & \frac{20}{9} \end{vmatrix}$$

$$D = (20)^3 |A|$$

$$\Rightarrow \frac{D}{|A|} = 8000$$

6. **Ans. (10.00)**

Sol.: $|A|^{16} = 4^8 \cdot 5^{16}$

$$|A| = 10$$

$$x + y + z = 10$$

7. **Ans. (3.00)**

Sol.: $\frac{1}{a_1 a_n} + \frac{1}{a_2 a_{n-1}} + \dots + \frac{1}{a_n a_1}$

$$= \frac{2}{a_1 + a_n} \left[\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n} \right]$$

$$\Rightarrow 1 - \tan \theta = \frac{1}{1 + \sin 2\theta} (1 + \tan \theta)$$

$$\Rightarrow (\cos \theta - \sin \theta) = \frac{\cos \theta + \sin \theta}{(\cos \theta + \sin \theta)^2}$$

$$\Rightarrow \cos 2\theta = 1 \Rightarrow \theta = k\pi$$

8. **Ans. (10.00)**

Sol. $S_{100} = \frac{0}{\binom{100}{C_0}^5} + \frac{1}{\binom{100}{C_1}^5} + \dots$ (i)

Also $S_{100} = \frac{100}{\binom{100}{C_0}^5} + \frac{99}{\binom{100}{C_1}^5} + \dots$ (ii)

$\therefore$ On adding (1) and (2), we get

$$2S_{100} = 100t_{100}$$

$$\Rightarrow \frac{t_{100}}{S_{100}} = \frac{2}{100}$$